Commodore C16 / Plus/4 — Chip Pinouts

MAJOR SOCKETED ICS · 7501 CPU · 7360 TED · 6551 · 6529 · 4164 RAM · TEST & REPAIR

DIP pinouts from the top, notch up, pin 1 top-left. See the **Legend** card for notation. TED & 7501 verified against the Commodore Plus/4 Technical Manual.

Legend — reading the pinouts (notation)

/NAME **Active LOW** — true / asserted when the pin is at logic **LOW** (~0 V). A leading **/** (datasheets often use an overline) marks these: e.g. `/RES /IRQ /CS /WE`.

NAME **Active HIGH / normal** — asserted when the pin is **HIGH** (~+5 V). No slash. `R/W` is a level: HIGH = read, LOW = write.

Colour key: **Vcc / Vdd = power** · **GND / Vss = ground** · **φ = clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

7360/8360 TED video/sound/MMU (48-DIP)

A2	1	48	A3
A1	2	47	A4
A0	3	46	A5
VCC	4	45	A6
CS0	5	44	A7
CS1	6	43	A8
R/W	7	42	A9
/IRQ	8	41	A10
MUX	9	40	A11
/RAS	10	39	A12
/CAS	11	38	A13
φOUT	12	37	A14
COLOR	13	36	A15
φIN	14	35	AEC
K0	15	34	BA
K1	16	33	SND
K2	17	32	DB7
K3	18	31	DB6
K4	19	30	DB5
K5	20	29	DB4
K6	21	28	DB3
K7	22	27	DB2
LUM	23	26	DB1
VSS	24	25	DB0

The heart of the 264 series: video, 2-voice sound, DRAM controller (/RAS /CAS MUX), ROM chip-selects (a simple MMU) and keyboard scan (K0–K7). Runs hot — the #1 failure. **Fails:** no/garbled picture, no sound, dead machine. Fit a heatsink.

7501/8501 CPU (6510 HMOS) (40-DIP)

φIN	1	40	/RES
RDY	2	39	R/W
/IRQ	3	38	DB0
AEC	4	37	DB1
VCC	5	36	DB2
A0	6	35	DB3
A1	7	34	DB4
A2	8	33	DB5
A3	9	32	DB6
A4	10	31	DB7
A5	11	30	P0
A6	12	29	P1
A7	13	28	P2
A8	14	27	P3
A9	15	26	P4
A10	16	25	P6
A11	17	24	P7
A12	18	23	GATE
A13	19	22	A15
GND	20	21	A14

HMOS 6510 variant @ 0.89/1.76 MHz (dual-speed). 7-bit port (P0–P4,P6,P7) drives cassette + serial bus. GATE holds R/W off until /RAS. 8501 is identical. **Fails:** dead machine, no boot.

6551 ACIA / UART (28-DIP)

Vss	1	28	R/W
CS0	2	27	φ2
/CS1	3	26	/IRQ
/RES	4	25	D7
RxC	5	24	D6
XTLI	6	23	D5
XTLO	7	22	D4
/RTS	8	21	D3
/CTS	9	20	D2
TxD	10	19	D1
/DTR	11	18	D0
RxD	12	17	/DSR
RS0	13	16	/DCD
RS1	14	15	Vdd

Plus/4 only: RS-232 user port up to 19,200 bps. Vdd is on pin 15, not 28. Any 6551/65C51 works. **Fails:** no/garbled RS-232.

6529 single port I/O (20-DIP)

R/W	1	20	Vdd
P0	2	19	/CS
P1	3	18	D0
P2	4	17	D1
P3	5	16	D2
P4	6	15	D3
P5	7	14	D4
P6	8	13	D5
P7	9	12	D6
Vss	10	11	D7

Single 8-bit latching port. Plus/4 uses two: one scans the keyboard columns, one is the user port. Write when /CS & R/W low. **Fails:** dead keyboard row or user port.

4164 DRAM 64K×1 (16-DIP)

NC	1	16	Vss
DIN	2	15	/CAS
/WE	3	14	DOUT
/RAS	4	13	A6
A0	5	12	A3
A2	6	11	A4
A1	7	10	A5
Vcc	8	9	A7

Eight 64K×1 DRAMs = 64KB. TED generates /RAS / CAS and MUX. **Fails:** garbled screen, won't boot, or random crashes — swap suspect 4164s.

Memory & ROM (RAM/ROM)

RAM	8× 4164 = 64KB
ROM	2× 23128 (16K×8): BASIC 3.5 + KERNAL
C16/116	16KB RAM, no user port
Plus/4	64KB + 3-plus-1 ROM + ACIA

23128 ROMs are 28-pin mask ROMs (27128-style). TED selects ROM via CS0 ($8000-BFFF) / CS1 ($C000-FFFF).

Test & Repair (tips)

Hot chip / no video	TED 7360 (add heatsink)
No boot	7501 / clock
Garbled RAM	4164
Dead keyboard	6529 / TED
No RS-232	6551 (Plus/4)
"Black screen"	check 5V + 7501 reset

TED's high failure rate is heat-driven. Only source of a replacement TED/7501 is another 264-series board, so socket them.